Medical Cannabis and Cannabinoids

Legal and Regulatory Aspects

Med Cannabis Cannabinoids 2025;8:58–64
DOI: 10.1159/000546264

Received: November 24, 2024
Accepted: May 1, 2025
Published online: May 14, 2025

Physicians' Knowledge, Attitudes, and Perceptions about Medical Cannabis in the United States: A Scoping Review

Eleanor Yusupov[a] Stephanie Lopez[b] Maria A. Pino[a]

[a]Department of Clinical Specialties, New York Institute of Technology, College of Osteopathic Medicine, Old Westbury, NY, USA; [b]Medical Library, New York Institute of Technology, College of Osteopathic Medicine, Old Westbury, NY, USA

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

Keywords
Medical cannabis · Patients · Scoping review · General practice · Safety

Abstract
Introduction: As legalization of medical cannabis (MC) in the USA expands, there remains uncertainty in clinical guidance. Healthcare professionals remain unprepared to communicate to patients the therapeutic outcomes and possible adverse effects of MC utilization. There is limited training provided at all levels of medical education, even for professionals with many years of clinical practice. Additionally, there is minimal scientific research, which delays the development of evidence-based guidelines. ***Methods:*** This review followed established methodological approaches for scoping reviews according to PRISMA-ScR guidelines. Studies were included if they addressed the attitudes and beliefs of medical practitioners in the USA and were published after the year 2000. ***Results:*** There were forty-one studies from January 2013 to February 2025 included in the format of both electronic surveys and qualitative interviews. Participants included US physicians, other healthcare professionals, and medical trainees, representing multiple clinical specialties. Physicians reported lack of confidence in counseling patients or managing their use of MC. Oncologists, emergency medicine physicians, pain management specialists, and primary care physicians perceived that MC is beneficial for managing chronic pain, nausea, loss of appetite, depression, and other symptoms. Obstetric providers had unfavorable perceptions about perinatal use of MC. Physicians practicing in states where the drug has been legalized and those with greater years of practice were more comfortable recommending MC and counseling patients. ***Conclusions:*** US physicians and medical trainees perceived significant knowledge barriers to recommending MC and counseling patients on its therapeutic use. Implementing clear clinical practice guidelines, further education on these drugs in clinical curriculums, and enhancing continuing education offerings would improve prescriber confidence. Increased research could also assist medical professionals in appropriate clinical decision making.

© 2025 The Author(s).
Published by S. Karger AG, Basel

Introduction

Cannabis has been utilized for its medicinal properties for centuries as it consists of approximately 540 natural compounds [1]. The predominant psychotropic

karger@karger.com
www.karger.com/mca

© 2025 The Author(s).
Published by S. Karger AG, Basel

Karger
OPEN ACCESS

This article is licensed under the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC) (http://www.karger.com/Services/OpenAccessLicense). Usage and distribution for commercial purposes requires written permission.

Correspondence to:
Maria A. Pino, mpino@nyit.edu

component is Δ9-tetrahydrocannabinol (Δ9-THC), while the major non-psychoactive ingredient is cannabidiol (CBD). Both compounds act on the cannabinoid receptors, CB_1 and CB_2 [1].

In the USA, the current classification of marijuana by the Drug Enforcement Agency (DEA) remains at Schedule I by the Controlled Substances Act of 1970. This category is considered to possess high abuse potential, with no accepted medical indication. In 2023, The Department of Health and Human Services recommended that marijuana be moved to Schedule III [2], as there is the potential for therapeutic use of the drug including the management of cancer-related adverse effects, infections, skin and neurological disorders [3].

There are four FDA approved cannabinoids for therapeutic indications which can be obtained by a prescription [4]. The purified form of CBD, cannabidiol, is used for the management of seizures in patients with Lennox-Gastaut or Dravet syndrome. There are two forms of dronabinol that are both synthetic THC. One form is used to manage chemotherapy-induced nausea and vomiting, while the other has an added benefit of treating cachexia in AIDS patients. Nabilone is another synthetic THC, used for both nausea and neuropathic pain [4, 5]. The approved medications are continuously monitored and were marketed following a thorough examination of quality, efficacy, and safety. Currently, there is a lack of FDA approval for any other cannabis, cannabis-derived, or CBD products available on the market.

California first legalized medical cannabis (MC) in 1996 via the Compassionate Care Act [6]. Currently, forty-seven states, the District of Columbia, and three territories (Guam, Puerto Rico, US. Virgin Islands) allow for the use of cannabis for medical purposes [7]. Thirty-eight states, the three territories, and DC have legalized the use of MC through comprehensive programs [7], while fourteen states and two territories have a comprehensive medical-only program [7]. Lastly, nine states have medical programs that only allow for the use of CBD/low-THC products for qualifying medical condition(s) as defined by their state [7]. Thus, MC cannot be prescribed by a standard prescription; only recommended by prescribers for patients who present with a qualifying medical condition, with great variation between states.

In all states that have legalized MC, chronic pain is the most common qualifying condition reported by patients [8]. After the patient obtains a "medical marijuana card," the drug is administered through a state-licensed dispensary. In a 2023 Gallup poll, it was noted that 70% of the American public supports the legalization of marijuana for both medical and recreational use [9]. The mismatch between federal, state, and interstate guidelines, along with varied public opinion, presents challenges for the medical community.

Clinician guidance related to managing patients with MC remains limited. Although there is no written prescription for MC, providers must still be prepared to educate their patients on the indications and therapeutic mechanisms of MC, risks for adverse effects or possible drug interactions, while refuting false claims regarding use of the drug for medicinal or recreational purposes. It has been reported that physicians receive minimal training on cannabinoids in medical school, leaving them oftentimes unprepared to make informed patient decisions [10]. Other healthcare professionals also cited a low level of knowledge in making appropriate patient recommendations regarding MC [11]. Scientific research has also lagged in providing evidence-based information for using MC for the indications mentioned (inflammatory, psychiatric, and dermatological conditions) and for the management of pediatric, pregnant, and geriatric patients.

The aim of this review was to identify and describe, through a scoping review approach, the attitudes and perspectives of US physicians, medical students, and other healthcare professionals on recommending MC, with a focus on examining their educational experiences, while evaluating the potential barriers for integrating MC into patient care.

Methods

This study examined the degree of knowledge and experience, attitudes, along with MC recommendation practices among healthcare professionals and trainees. A scoping review was appropriate to evaluate these objectives since it can address research questions of a complex or heterogeneous nature [12]. In this review, we followed the reporting guidelines outlined by the Preferred Reporting Items for Systematic reviews and Meta-Analyses Extension for Scoping Review (PRISMA-ScR) [13]. Peer-reviewed articles addressing the attitudes and beliefs of medical practitioners on MC prescribing practices were searched systematically via Ovid MEDLINE, Scopus, Cochrane Library, PubMed, CINAHL, and ProQuest Psychology Database, with additional searches through the grey literature via medRxiv, and bioRxiv. These databases were searched

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

Table 1. Search terms utilized in this scoping review

	Concept 1	Concept 2	Concept 3	Concept 4
Key concepts	Physicians	Beliefs or policies around medical cannabis	Prescribing practices	Medical cannabis
Natural language terms	Physicians, doctors, osteopathic physicians, prescribers, psychiatrists, health workers, health professionals, health personnel	Attitude, beliefs, healthcare policies, policies, morality, morals, lifestyle, prejudice, health policy, views, laws, regulations, rules	Prescribing, prescriptions, medication administration, physician patterns, physician behaviors, medical order, medication order, off-label use	Medical cannabis, medical marijuana, marijuana, cannabis, cannabinoids, weed, THC, prescription THC, prescription cannabis, prescription marijuana
Controlled vocabulary terms	Physicians; health personnel	Attitude; social control policies; risk assessment; psychology; social; stereotyping	Prescriptions; drug therapy; pain management; patient care management; delivery of health care	Cannabis; medical marijuana; marijuana use; cannabinoids

from January 2013 to February 2025. The search strategy included search terms listed in Table 1.

The search strategy was adjusted for each database as necessary using database-specific logic and wildcards. During the initial searches, there were no restrictions to publication date, language, document type, or study type. Hand searching was also performed upon manual review of the references of included articles.

The inclusion and exclusion criteria along with the reasoning are outlined as follows: included studies were those that focused on attitudes and beliefs of medical professionals within the USA about MC regardless of the preparation of the medical drug. Exclusion criteria were those studies occurring prior to the year 2000 as MC was not legal nor commonplace, studies that inquired about other drugs, recreational use of cannabis, studies that focused on MC in patient populations under the age of 18, studies conducted outside the USA, and studies that focused on the patient or nonmedical professional perspective.

The articles were uploaded to Zotero and Rayyan for deduplication at which point the two subject matter experts screened the titles and abstracts of the articles based on inclusion and exclusion criteria with conflicts resolved through discussion. Preliminary scans of the academic literature allowed for the identification of some eligible articles. The subject matter experts identified primary research articles and research protocols that evaluated the perceptions of physicians, student physicians, and allied health professionals on MC and its uses both recreationally and clinically. Articles in the form of commentaries, editorials, letters to the editor, conference abstracts, and opinion pieces were also ineligible.

For data extraction, we developed a table that allowed the subject matter experts to evaluate each article and to identify any relevant information. The following data from each eligible article were summarized and extracted: author(s); year of publication; population/participants; research design; clinical indications; and evaluation of physicians', student physicians', and allied health professionals' perceptions of marijuana and MC. The data were summarized in table format with the descriptive data analyzed using content analysis. The subject matter experts reviewed the descriptive data and resolved any discrepancies through discussions. The findings were then organized by themes (online suppl. Table 1; for all online suppl. material, see https://doi.org/10.1159/000546264).

Results

There were forty-one articles ultimately selected for inclusion in this review.

Types of Studies

Most of the reviewed studies were administered as electronic surveys, while three were qualitative interviews [14–16]. One published work was based on a written test administered before a CME program on MC [17].

Recommendations of MC by Various Medical Specialties

A summary of healthcare professionals included in this review and their perceptions of MC is provided in online supplementary Table 2. Multiple researchers focused on the attitudes and beliefs of physicians and those in

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

training regarding the various indications of MC including: Ananth et al. [18]; Novak et al. [19]; Azizoddin et al. [20]; Braun et al. [21]; Luba et al. [22]; Narouze et al. [23]; Cooke et al. [24]; Starrels et al. [25]; Phillips et al. [26]; Robinson et al. [27]; Yeroushalmi et al. [28]; Holland et al. [15]; Young-Wolff et al. [29]; Takakuwa and Schears [30]; and Kondrad and Reid [31].

Dermatologists

Perceptions of dermatologists on MC use was the subject of two studies conducted in 2018 and 2020 via online surveys, with most physicians recommending MC as topical preparations for the management of dermatological conditions [27, 28]. The most common dermatological indications cited were inflammatory and pruritic disorders, such as psoriasis and atopic dermatitis [28].

Emergency and Family Medicine Physicians

Most emergency medicine physicians believed MC was valuable, especially for chronic pain and nausea/vomiting management. However, use of MC was unfavorable in pregnant and pediatric populations [30]. Additionally, family medicine physicians were not convinced of MC benefits and believe its use poses risks [31]. However, cancer and cachexia were listed as the most frequent reasons for possible MC recommendation [31]. Nearly, all physicians who responded to the surveys agreed on the need for further medical education about MC [31].

Oncologists

In a survey of pediatric oncologists, most of those who responded believed that the use of MC was appropriate in the treatment of children with cancer [18], while less than half of medical oncologists in the large national survey felt that it was beneficial in cancer patients over 65 years of age [20]. Oncologists also believed that MC were most useful in treating cancer-related nausea, severe pain, loss of appetite, and depression [19, 21]. Oncologists in states eligible to recommend MC support its utilization in the palliative care setting [18].

Other Specialty Perspectives

Most physicians believe that MC is harmful for patients with congenital heart disease and most pediatricians overall would not recommend its use [32]. Neurologists still perceive a stigma regarding the recommendation of MC for epilepsy [33]. Obstetric providers were more likely to have unfavorable perceptions of perinatal cannabis use [15, 29, 34, 35]. The lack of definitive information on the risks of MC in perinatal use was a barrier to patient counseling [15]. When counseling, the physicians described raising the possibility of legal consequences or involvement of child protective services as their primary approach [15]. Psychiatrists and psychologists were more likely to discourage versus encourage prenatal cannabis use relative to obstetrician/gynecologists [29]. Pain management and primary care physicians believed that MC would be useful in treating cancer-related nausea, severe pain, loss of appetite, depression [24, 36], as well as chronic non-cancer pain [23, 24].

Palliative Care Physicians

Palliative care and hospice physicians perceived cannabis as helpful in treating pain, nausea, appetite loss, sleep problems, irritability, agitation, emotional suffering, and end-of-life care more broadly [22, 37]. However, fewer than half reported having recommended cannabis to patients in the past [22].

Trends of Physician Attitudes and Beliefs in Different Geographical Regions

Several studies found that physicians had more favorable opinions of MC or were more likely to recommend it in states where MC has been legalized [19, 26, 27, 38]. Physiatrists were more likely to have obtained Certificate to Recommend marijuana required in Ohio than family medicine physicians, internists or pediatricians [39]. Physiatrists are physicians who specialize in managing chronic pain and other injuries. A nationally representative survey of oncologists revealed that practicing physicians in the Western region of the USA were significantly more likely to discuss MC with their patients than those practicing in rural areas or the South [21, 32, 38].

Additional Trends in MC Recommendations

Older physicians with greater experience in their medical practice [28] were more comfortable discussing MC with patients and more likely to recommend its use than younger physicians [27]. DOs were more likely than MDs to recommend MC in the state of Ohio [39]. Some respondents noted the importance of the setting they were practicing (academic institution vs. private practice), along with the years of experience [19, 32]. Those with minimal knowledge at identifying MC adverse effects or effective doses were not comfortable incorporating MC into their medical practice [11].

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

Physician Knowledge about MC and Preparedness to Prescribe and Counsel

Physicians representing different specialties from across the USA recognized gaps in their knowledge about recommending MC to their patients [11, 17, 19, 21, 27–29, 38, 40–43]. Self-perceived knowledge deficits affected physicians' preparedness to prescribe MC and counsel patients [11, 17, 20, 28, 36, 38, 43]. Most of those who responded were unsure of the risk vs. benefits of cannabinoid use and there was a lack of knowledge of cannabinoid components [33, 42–44]. Therefore, there was a need to be educated on the basic pharmacology of these compounds [17, 27, 40].

Physician Attitudes toward Further Education on MC

Physicians expressed the need for postgraduate education on MC in different formats, including CME (webinars and conference presentations), along with clearer state clinical practice guidelines [16, 18–20, 30–34, 36, 38, 40, 45]. Further basic science and clinical research involving MC would be beneficial, especially as they apply to appropriate patient care [25, 27, 30, 40, 46].

Physicians-in-Training and Academic Administration Perspectives of MC Role in Medical Education

There was a lack of education and preparedness of physicians-in-training to prescribe or discuss MC with patients [10, 42]. Knowledge of cannabis was moderate, along with the concern of cannabis negative effects [47, 48]. While medical students favored legalization, they still endorsed medical use more than recreational use [47]. Most medical students expressed that cannabis has a role in medicine and should be reclassified as a noncontrolled substance [47]. Medical educators, curriculum deans as well as residents and fellows supported the inclusion of training on MC into medical school curriculums [10, 33, 47].

Discussion

The use of MC has gained much public and political support in the USA. As many states have eased restrictions on cannabinoid access for medical purposes, healthcare professionals must adapt to changes. Namely, these include whether to recommend MC to their patients, and how to counsel them effectively. This scoping review aimed to examine which barriers persist in healthcare providers' decisions to recommend MC and how they believe these challenges can be mitigated.

The most common barriers identified by physicians and medical trainees is an insufficient knowledge of MC indications, formulations, prescribing information, drug interactions, and adverse effects [11, 14, 16–19, 21, 24, 25, 27, 30, 33, 40, 44, 49]. Physicians welcome clearer clinical practice guidelines when recommending MC, especially since the legal status of marijuana remains inconsistent between states [14, 17, 18, 24, 41, 43, 49]. Continuing medical education courses can be utilized to increase physicians' understanding of up-to-date MC guidelines [25, 50, 51]. Credentialing organizations, along with state and specialty boards, already require physicians and other licensed healthcare professions to periodically complete CME courses to maintain licensure. Expanding CME offerings to teach MC prescribing and pharmacological principles would be a practical way to better equip practicing physicians with needed knowledge and confidence [15, 19, 31–34, 38, 40, 45]. Medical conferences are also an appropriate platform for relating information to healthcare professionals. Current research on this expanding topic can be presented to both healthcare professionals and those training for careers in patient care. Research can help inform expanded indications of MC, mechanisms of its use, safety and pharmacokinetic parameters, and evidence-based guidelines. As research is integral to the future of MC acceptance, states should continue to offer financial incentives for institutions engaged in studying MC. Medical educators continue to search for appropriate methods to incorporate MC into their curriculum. Students, residents, and fellows remain minimally prepared on this topic, both on the pharmacological principles and the ability to provide guidance to patients who have questions on MC [10, 38, 42, 48]. Of great importance, is how to effectively communicate to patients the differences between using marijuana for medical purposes versus recreationally. While it may make sense to integrate the learning of MC throughout all years of a student's medical education, it must be considered that the training received in the clinical years also depends on the knowledge, comfort, and confidence level of the preceptor on rotations. This again enforces the need for appropriate continuing education methods for everyone educating the next generation of healthcare professionals.

Another obstacle to provider recommendation of MC involves some providers' concerns about misuse of cannabinoid access for recreational purposes. Cannabinoids can be highly abused recreational substances that may be criminal to those individuals who have them in possession [18, 24]. There may be concerns if patients are in fact using cannabinoids medically or are seeking the substance from a healthcare provider for recreational use. Among those physicians that most support the use of MC

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

are oncologists and hospice care providers, who utilize cannabinoids for the management of chemotherapy-induced nausea and vomiting, poor appetite, and pain [19, 21, 37]. These physicians may have an increased confidence in MC benefits and understand the appropriate monitoring of adverse effects due to their experiences in their specific patient population. This highlights that further research on cannabinoid use for medical purposes and clearer legal guidelines can be helpful in increasing provider confidence [49]. In conclusion, effectively training current and future healthcare professionals on MC is a necessary step toward improving patient care and reducing potential stigma associated with medical versus recreational cannabinoid use.

Statement of Ethics

Not applicable.

Conflict of Interest Statement

There are no competing interests.

Funding Sources

The authors received no funding for this work.

Author Contributions

S.L. wrote the methods and assembled the tables/references, helped obtain the articles. E.Y. and M.A.P. wrote the abstract, introduction, results, and discussion, reviewed articles, and gave idea. All authors reviewed the manuscript.

Data Availability Statement

This was a scoping review; no data were collected.

References

1 Amin MR, Ali DW. Pharmacology of medical cannabis. In: Bukiya AN, editor. Recent advances in cannabinoid physiology and pathology. Springer International Publishing; 2019. p. 151–65.
2 HHS recommends that marijuana be reclassified as Schedule III controlled substance. Available from: https://www.maynardnexsen.com/publication-hhs-recommends-that-marijuana-be-reclassified-as-schedule-iii-controlled-substance (Accessed March 25, 2025).
3 Pagano C, Navarra G, Coppola L, Avilia G, Bifulco M, Laezza C. Cannabinoids: therapeutic use in clinical practice. Int J Mol Sci. 2022;23(6):3344. https://doi.org/10.3390/ijms23063344
4 FDA and cannabis: research and drug approval process. FDA; 2023. https://www.fda.gov/news-events/public-health-focus/fda-and-cannabis-research-and-drug-approval-process (Accessed March 25, 2025).
5 About cannabis: Cannabis and public health; 2024. https://www.cdc.gov/cannabis/about/index.html (Accessed March 25, 2025).
6 California's cannabis laws. Department of Cannabis Control. https://cannabis.ca.gov/cannabis-laws/laws-and-regulations/ (Accessed March 25, 2025).
7 CDC. State medical cannabis laws. Cannabis and Public Health; 2024. Available from: https://www.cdc.gov/cannabis/about/state-medical-cannabis-laws.html (Accessed March 25, 2025).
8 Boehnke KF, Gangopadhyay S, Clauw DJ, Haffajee RL. Qualifying conditions of medical cannabis license holders in the United States. Health Aff. 2019;38(2):295–302. https://doi.org/10.1377/hlthaff.2018.05266
9 Shah S, Chapekis A. Most Americans now live in a legal marijuana state – and most have at least one dispensary in their county.Pew Research Center; 2024. Available from: https://www.pewresearch.org/short-reads/2024/02/29/most-americans-now-live-in-a-legal-marijuana-state-and-most-have-at-least-one-dispensary-in-their-county/ (Accessed March 25, 2025).
10 Evanoff AB, Quan T, Dufault C, Awad M, Bierut LJ. Physicians-in-training are not prepared to prescribe medical marijuana. Drug Alcohol Depend. 2017;180:151–5. https://doi.org/10.1016/j.drugalcdep.2017.08.010
11 Kruger DJ, Mokbel MA, Clauw DJ, Boehnke KF. Assessing health care providers' knowledge of medical cannabis. Cannabis Cannabinoid Res. 2022;7(4):501–7. https://doi.org/10.1089/can.2021.0032
12 Peters MDJ, Godfrey CM, Khalil H, McInerney P, Parker D, Soares CB. Guidance for conducting systematic scoping reviews. JBI Evid Implement. 2015;13(3):141–6. https://doi.org/10.1097/XEB.0000000000000050
13 Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med. 2018;169(7):467–73. https://doi.org/10.7326/M18-0850
14 Boullosa CC, Vaddiparti K, Lopez-Quintero C, Varma DS, L Cook R. Clinical decision making by medical marijuana physicians in Florida: a qualitative assessment. Subst Use Misuse. 2022;57(14):2042–52. https://doi.org/10.1080/10826084.2022.2129995
15 Holland CL, Nkumsah MA, Morrison P, Tarr JA, Rubio D, Rodriguez KL, et al. "Anything above marijuana takes priority": obstetric providers' attitudes and counseling strategies regarding perinatal marijuana use. Patient Educ Couns. 2016;99(9):1446–51. https://doi.org/10.1016/j.pec.2016.06.003
16 Bascombe TMS, Scott KN, Ballard D, Smith SA, Thompson W, Berg CJ. Primary healthcare provider knowledge, beliefs and clinic-based practices regarding alternative tobacco products and marijuana: a qualitative study. Health Educ Res. 2016;31(3):375–83. https://doi.org/10.1093/her/cyv103
17 Rice J, Hildebrand A, Waslo CS, Cameron MH, Jones KD. Cannabis for medical purposes: a cross-sectional analysis of health care professionals' knowledge. J Am Assoc Nurse Pract. 2021;34(1):100–6. https://doi.org/10.1097/JXX.0000000000000590
18 Ananth P, Ma C, Al-Sayegh H, Kroon L, Klein V, Wharton C, et al. Provider perspectives on use of medical marijuana in children with cancer. Pediatrics. 2018;141(1):e20170559. https://doi.org/10.1542/peds.2017-0559
19 Novak J, Liu J, Zou X, Abuali T, Vazquez J, Kalash R, et al. Radiation oncologist perceptions of therapeutic cannabis use among cancer patients. Support Care Cancer. 2021;29(10):5991–7. https://doi.org/10.1007/s00520-021-06160-6
20 Azizoddin DR, Enzinger AC, Wright AA, Yusufov M, Tulsky JA, Campbell EG, et al. Oncologists' perspectives on medical marijuana use by older adults. J Geriatr Oncol. 2020;11(6):1034–7. https://doi.org/10.1016/j.jgo.2019.12.014

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025

21 Braun IM, Wright A, Peteet J, Meyer FL, Yuppa DP, Bolcic-Jankovic D, et al. Medical oncologists' beliefs, practices, and knowledge regarding marijuana used therapeutically: a nationally representative survey study. J Clin Oncol. 2018;36(19):1957–62. https://doi.org/10.1200/JCO.2017.76.1221
22 Luba R, Earleywine M, Farmer S, Slavin M. Cannabis in end-of-life care: examining attitudes and practices of palliative care providers. J Psychoactive Drugs. 2018;50(4):348–54. https://doi.org/10.1080/02791072.2018.1462543
23 Narouze S, Hakim SM, Kohan L, Adams D, Souza D. Medical cannabis attitudes and beliefs among pain physicians. Reg Anesth Pain Med. 2020;45(11):917–9. https://doi.org/10.1136/rapm-2020-101658
24 Cooke AC, Knight KR, Miaskowski C. Patients' and clinicians' perspectives of co-use of cannabis and opioids for chronic non-cancer pain management in primary care. Int J Drug Pol. 2019;63:23–8. https://doi.org/10.1016/j.drugpo.2018.09.002
25 Starrels JL, Young SR, Azari SS, Becker WC, Jennifer Edelman E, Liebschutz JM, et al. Disagreement and uncertainty among experts about how to respond to marijuana use in patients on long-term opioids for chronic pain: results of a delphi study. Pain Med. 2020;21(2):247–54. https://doi.org/10.1093/pm/pnz153
26 Phillips KA, Thrush PT, Lal AK, Kindel SJ, Castleberry C, Sparks J, et al. Marijuana in pediatric and adult congenital heart disease heart transplant listing: a survey of provider practices and attitudes. Pediatr Transpl. 2020;24(2):e13640–5. https://doi.org/10.1111/petr.13640
27 Robinson E, Murphy E, Friedman A. Knowledge, attitudes, and perceptions of cannabinoids in the dermatology community. J Drugs Dermatol. 2018;17(12):1273–8.
28 Yeroushalmi S, Nelson K, Sparks A, Friedman A. Perceptions and recommendation behaviors of dermatologists for medical cannabis: a pilot survey. Complement Ther Med. 2020;55:102552. https://doi.org/10.1016/j.ctim.2020.102552
29 Young-Wolff KC, Gali K, Sarovar V, Rutledge GW, Prochaska JJ. Women's questions about perinatal cannabis use and health care providers' responses. J Womens Health. 2020;29(7):919–26. https://doi.org/10.1089/jwh.2019.8112
30 Takakuwa KM, Schears RM. Emergency medicine physicians would prefer using cannabis over opioids for first-line treatment of a medical condition if provided with medical evidence: a national survey. Cureus. 2021;13(11):e19641. https://doi.org/10.7759/cureus.19641
31 Kondrad E, Reid A. Colorado family physicians' attitudes toward medical marijuana. J Am Board Fam Med. 2013;26(1):52–60. https://doi.org/10.3122/jabfm.2013.01.120089
32 Ross B, Trojian T, Cushman DM. Physician perceptions of cannabidiol (CBD) and cannabis in sports medicine and performance. Transl Sports Med. 2023;2023:8824466. https://doi.org/10.1155/2023/8824466
33 Szaflarski M, McGoldrick P, Currens L, Blodgett D, Land H, Szaflarski JP, et al. Attitudes and knowledge about cannabis and cannabis-based therapies among US neurologists, nurses, and pharmacists. Epilepsy Behav. 2020;109:107102. https://doi.org/10.1016/j.yebeh.2020.107102
34 McCall D, Rapoza S, Liverman W, Carter J. Nurses' attitudes toward patients who use cannabis: does legal status or care setting matter? J Addict Nurs. 2023;34(3):178–85. https://doi.org/10.1097/JAN.0000000000000528
35 Kitsantas P, Pursell SR. Are health care providers caring for pregnant and postpartum women ready to confront the perinatal cannabis use challenge? Am J Perinatol. 2024;41(S 01):e3249–54. https://doi.org/10.1055/s-0043-1777669
36 Philpot LM, Ebbert JO, Hurt RT. A survey of the attitudes, beliefs and knowledge about medical cannabis among primary care providers. BMC Fam Pract. 2019;20(1):17. https://doi.org/10.1186/s12875-019-0906-y
37 Bobitt J, Berryman K, Weaver FM. VA providers' perceptions of cannabis use policies in a legalized and nonlegalized state. Cannabis Cannabinoid Res. 2025;10(1):e176–83. https://doi.org/10.1089/can.2023.0276
38 Carlini BH, Garrett SB, Carter GT. Medicinal cannabis: a survey among health care providers in Washington state. Am J Hosp Palliat Med. 2017;34(1):85–91. https://doi.org/10.1177/1049909115604669
39 Leeds FS, Levinthal RK, Alexander MT, Crawford TN. The distribution of Ohio's Certificates to Recommend: who will "prescribe" medical marijuana? J Cannabis Res. 2020;2(1):11. https://doi.org/10.1186/s42238-020-00019-z
40 Muirhead CA, Palmrose W, Condren M, Rotolo SM, Pettit RS, Gill M, et al. A clinician survey of use assessment, documentation, and education about cannabis in persons with cystic fibrosis. Pediatr Pulmonol. 2021;56(9):2879–87. https://doi.org/10.1002/ppul.25563
41 Sharma P, Wilfahrt RP, Lackore K, Hammond CJ. Differences in cannabidiol-related attitudes and practice behaviors between U.S. Primary care physicians practicing in a single health care system across states with and without marijuana legalization. Cannabis Cannabinoid Res. 2024;9(4):1038–47. https://doi.org/10.1089/can.2023.0011
42 Sherry DR, Patell R, Han HJ, Dodge LE, Braun IM, Buss MK. Palliative medicine fellows' discussions, perceptions, and training regarding medical cannabis. J Pain Symptom Manage. 2024;67(6):471–7.e6. https://doi.org/10.1016/j.jpainsymman.2024.02.562
43 Brooks E, Gundersen DC, Flynn E, Brooks-Russell A, Bull S. The clinical implications of legalizing marijuana: are physician and non-physician providers prepared? Addict Behav. 2017;72:1–7. https://doi.org/10.1016/j.addbeh.2017.03.007
44 Griffith ST, Conrow KD, Go M, McEntee ML, Daniulaityte R, Nadesan MH, et al. Cannabis use in Parkinson's disease: patient access to medical cannabis and physician perspective on product safety. Neurotoxicology. 2024;103:198–205. https://doi.org/10.1016/j.neuro.2024.05.008
45 Bui K, Worster B, Ryan J, Pant S, Kelly EL. Physicians' perspectives on their training for and role within Pennsylvania's medical cannabis program. J Am Board Fam Med. 2023;36(4):670–81. https://doi.org/10.3122/jabfm.2022.220304R1
46 Kruger DJ, Gerlach J, Kruger JS, Mokbel MA, Clauw DJ, Boehnke KF. Physicians' attitudes and practices regarding cannabis and recommending medical cannabis use. Cannabis Cannabinoid Res. 2024;9(4):e1048–55. https://doi.org/10.1089/can.2022.0324
47 Jacobs RJ, Kane MN, Caballero J, Jacobs RJ, Kane MN, Caballero J. Predictors of medical students' perceptions about medical cannabis. Cureus. 2022;14(4):e24390. https://doi.org/10.7759/cureus.24390
48 Cassone C, Rissmiller P. NP student knowledge and perceptions of medical marijuana. Nurse Pract. 2024;49(9):1–8. https://doi.org/10.1097/01.NPR.0000000000000223
49 Zeiger JS, Silvers WS, Zeiger RS. Comparison of cannabis knowledge, attitudes, and practice between allergists and patients with allergies and asthma. Ann Allergy Asthma Immunol. 2024;133(3):344–5. https://doi.org/10.1016/j.anai.2024.06.001
50 Elbready AW, Warner-Maron I, Glicksman A, Peterson AM. Primary care providers' communication about medical cannabis with older adults: a cross-sectional survey. J Prim Care Community Health. 2024;15:21501319241295922. https://doi.org/10.1177/21501319241295922
51 Meyers S, Gant K, Burmeister M. Assessment of education in a community hospital on healthcare providers' knowledge of and attitudes toward medical marijuana. Curr Pharm Teach Learn. 2024;16(6):396–403. https://doi.org/10.1016/j.cptl.2024.03.007

Downloaded from http://karger.com/mca/article-pdf/8/1/58/4376331/000546264.pdf by guest on 17 July 2025